

Mass Transfer - 2

Mass Transfer Fundamentals

BSL

Fick's Law $N_A = x_A (N_A + N_B) - D_{AB} \nabla C_A$

$\uparrow$
 Molar Flux

$\uparrow$
 Bulk Flow

$\uparrow$
 Diffusion

$D_{AB} \left(\frac{\partial C_A}{\partial x} + \frac{\partial C_A}{\partial y} + \frac{\partial C_A}{\partial z} \right)$

Species Balance Equation in Cartesian Coordinate System

$$\left(\frac{\partial C_A}{\partial t} + v_x \frac{\partial C_A}{\partial x} + v_y \frac{\partial C_A}{\partial y} + v_z \frac{\partial C_A}{\partial z} \right) = D_{AB} \left[\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2} \right] + R_A$$

$\uparrow$
 Transient,
 $\downarrow$

$\uparrow$
 Convection,
 $\downarrow$

$\uparrow$
 Conduction/Diffusion,
 $\downarrow$

$\uparrow$
 Generation
 $\downarrow$

$$\rho \hat{C}_p \left(\frac{\partial T}{\partial t} + v_x \frac{\partial T}{\partial x} + v_y \frac{\partial T}{\partial y} + v_z \frac{\partial T}{\partial z} \right) = k \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] + \mu \phi_v + \dot{Q}$$

Observations

$$\rho \left(\frac{\partial C_A}{\partial t} + v_x \frac{\partial C_A}{\partial x} + v_y \frac{\partial C_A}{\partial y} + v_z \frac{\partial C_A}{\partial z} \right) = D_{AB} \left[\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2} \right] + R_A$$

Homogeneous reaction term appears in the governing equation

Heterogeneous reaction appears as a boundary condition.

Possible boundary conditions

1. Concentration specified at a point e.g. solubility, equilibrium etc.
2. At the catalyst surface, diffusive flux to the surface must be equal to the rate of reaction

$$- D_{AB} \frac{dC_A}{dx} \Big|_{x=\text{cat sur}} = - R_A //$$

// refers to heterogeneous reaction

Boundary conditions contd.

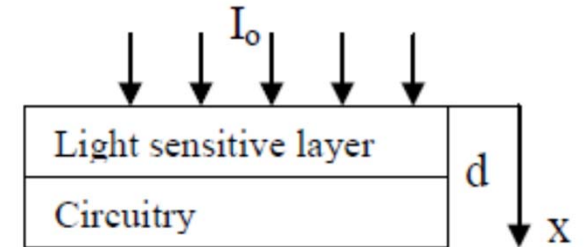
3. Equality of concentration at the interface $C_{A, s1} = C_{A, s2}$

4. Impervious surface $dC/dx = 0$ at specified location

5. Conductive flux to the solid-fluid interface is equal to the convective flux away from the interface, $-D_{AB} \left. \frac{dC_A}{dx} \right|_{x=L} = h_m (C_{As} - C_{A\infty})$

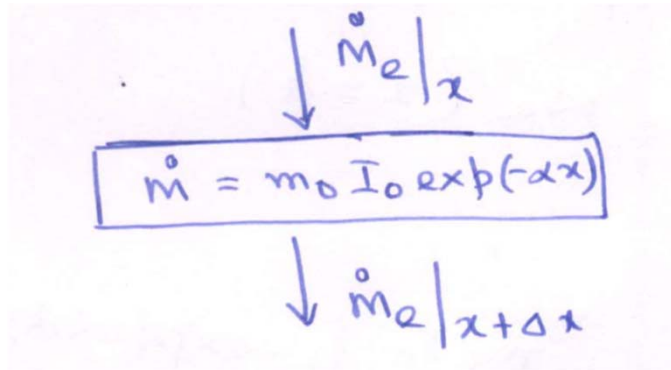
In a photodiode, as shown in the figure (d being the thickness of the sensitive layer), electrons are generated at a rate, $\dot{M}$ which are directly proportional to the incident light intensity. The intensity varies exponentially with depth as shown below

$$I = I_0 \exp(-\alpha x) \quad \text{and} \quad \dot{M} \text{ (mol / m}^3 \text{ s)} = m_0 I$$



The electrons diffuse toward the circuitry (the generated electrons flow through the interface at $x=d$) and a signal is generated that is proportional to the electron flux at d.

- (a) Evaluate an expression for the concentration profile of electrons, C_e , in the light sensitive layer. The flow of electrons is governed by a law similar to Fick's law. The diffusivity of electrons is D_{es} .
- (b) All photodiodes have a dark current, I_d , due to electrons formed by random thermal means, that is basically noise. The dark current density is proportional to the volume of light sensitive material, $I_d = I_{no} V$, where V is the volume and I_{no} is a constant. Develop an expression for the signal to noise ratio.



IN – OUT + Generation = Accumulation

$$\dot{M}_e(x) - \dot{M}_e(x + \Delta x) + \dot{M} A_c \Delta x = 0$$

$$-\frac{d \dot{M}_e}{dx} + \dot{M} A_c = 0 \quad (1)$$

Substituting Fick's law for the molar flow rate of electrons

$$\dot{M}_e = A_c N_c = -A_c D_{es} \frac{dc_e}{dx}$$

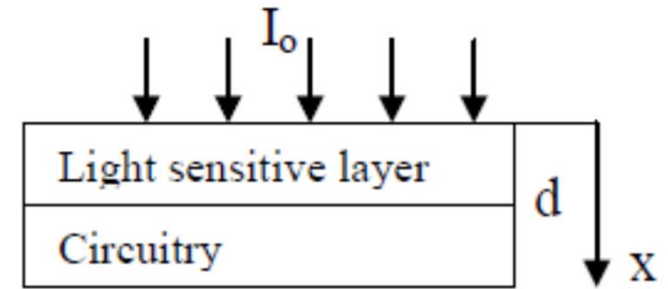
Therefore Eq. (1) becomes

$$D_{es} \frac{d^2 C_e}{dx^2} + m_0 I_0 \exp(-\alpha x) = 0$$

$$\frac{dC}{dx} = \frac{m_0 I_0}{\alpha D_{es}} e^{-\alpha x} + a_1$$

$$C = -\frac{m_0 I_0}{\alpha^2 D_{es}} e^{-\alpha x} + a_1 x + a_2$$

Boundary Conditions



i) No electrons at $x = 0$

ii) All generated electrons must flow through the interface at $x = d$

$$x = 0, C_e = 0 \rightarrow a_2 = \frac{m_o I_o}{\alpha^2 D_{es}}$$

$$x = d, M_e = A_c \int_0^d M dx \Rightarrow -D_{es} A_c \left. \frac{dC_A}{dx} \right|_d = A_c \int_0^d m_o I_o e^{-\alpha x} dx$$

$$-D_{es} \left[\frac{m_o I_o}{\alpha D_{es}} e^{-\alpha x} + a_1 \right]_{x=d} = \left[\frac{m_o I_o}{\alpha} e^{-\alpha x} \right]_0^d$$

$$\Rightarrow a_1 = -\frac{m_o I_o}{\alpha D_{es}}$$

$$C_e = \frac{m_o I_o}{\alpha^2 D_{es}} [1 - \exp(-\alpha x)] - \frac{m_o I_o}{\alpha D_{es}}$$

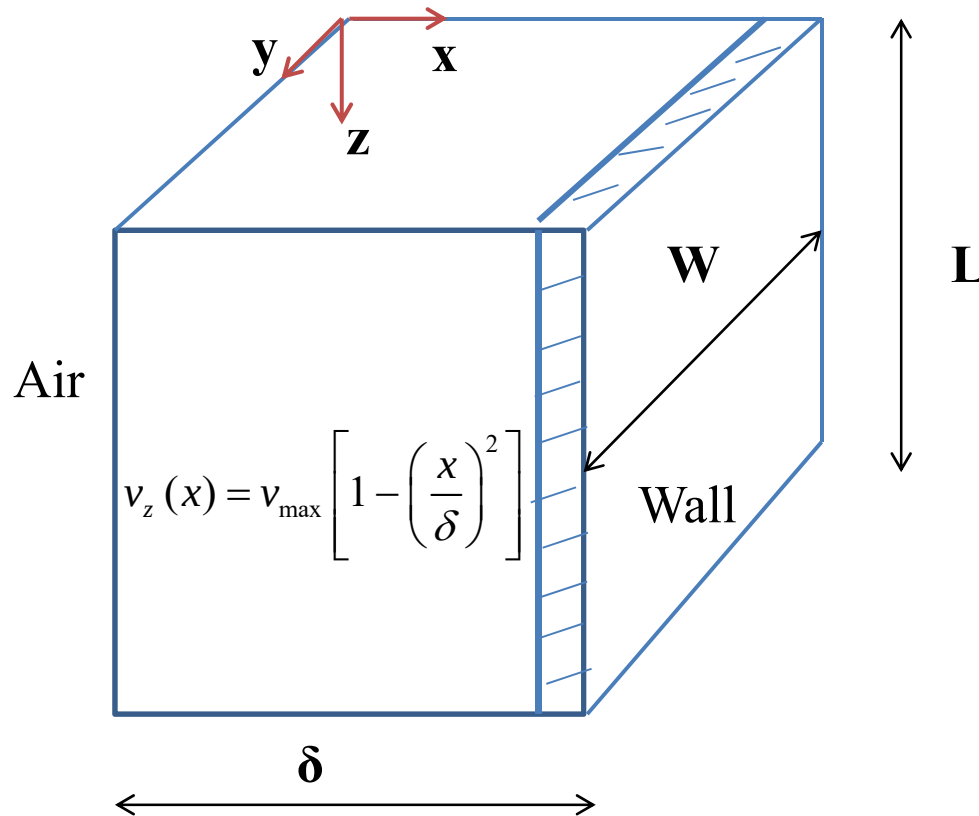
The signal we receive is proportional to the flow of electrons from the active layer.
The noise is given by the dark current density times the volume of the active layer.

$$Signal = \dot{M}_e(x = d) = A_c N_e(x = d)$$

$$Noise = \dot{I}_{no} A_c d$$

$$\frac{Signal}{Noise} = \frac{N_e(x = d)}{I_{no} d} = \frac{m_o I_o [1 - \exp(-\alpha d)]}{I_{no} d}$$

Diffusion into a falling Film - Forced Convection Mass Transfer



Absorption of gas A by a laminar falling film of liquid B

- (i) A is only slightly soluble in B
- (ii) Diffusion of A in B takes place very slowly, i.e. A does not penetrate very far into B – penetration distance small compared to the film thickness

$$\left(\cancel{\frac{\partial C_A}{\partial t}} + \cancel{v_x} \frac{\partial C_A}{\partial x} + \cancel{v_y} \frac{\partial C_A}{\partial y} + v_z \frac{\partial C_A}{\partial z} \right) = D_{AB} \left[\frac{\partial^2 C_A}{\partial x^2} + \cancel{\frac{\partial^2 C_A}{\partial y^2}} + \cancel{\frac{\partial^2 C_A}{\partial z^2}} \right] + \cancel{R_A}$$

$$\left(v_z \frac{\partial C_A}{\partial z} \right) = D_{AB} \left[\frac{\partial^2 C_A}{\partial x^2} \right]$$

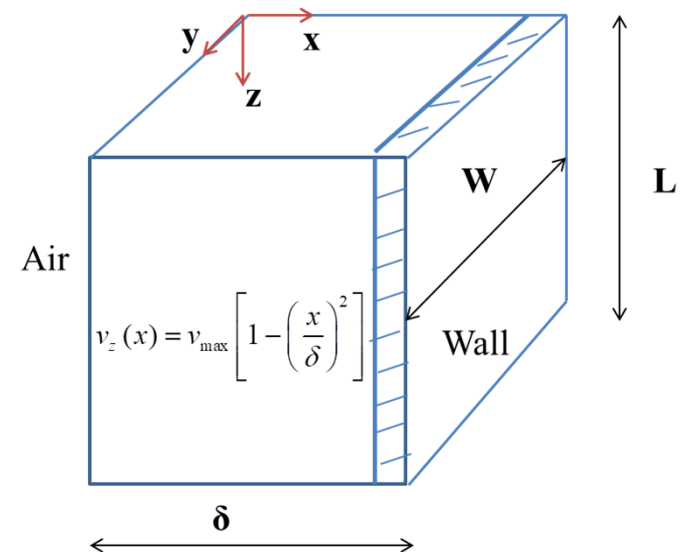
$$v_{\max} \left[1 - \left(\frac{x}{\delta} \right)^2 \right] \frac{\partial C_A}{\partial z} = D_{AB} \frac{\partial^2 C_A}{\partial x^2}$$

As the penetration distance of A in B is small (slow diffusion, small length etc.), to the diffusing A molecules the entire liquid film will appear as if travelling with $v_{\max}$

$$v_{\max} \frac{\partial C_A}{\partial z} = D_{AB} \frac{\partial^2 C_A}{\partial x^2}$$

BC

1. At $z = 0$, $C_A = 0$
2. At $x = 0$, $C_A = C_{A0}$
3. At $x = \delta$, $\partial C_A / \partial x = 0$



BC

$$v_{\max} \frac{\partial C_A}{\partial z} = D_{AB} \frac{\partial^2 C_A}{\partial x^2}$$

1. At $z = 0$, $C_A = 0$

2. At $x = 0$, $C_A = C_{AO}$

3. At $x = \delta$, $\partial C_A / \partial z = 0$

Since the film thickness is large as compared to the penetration distance the 3rd BC may be written as

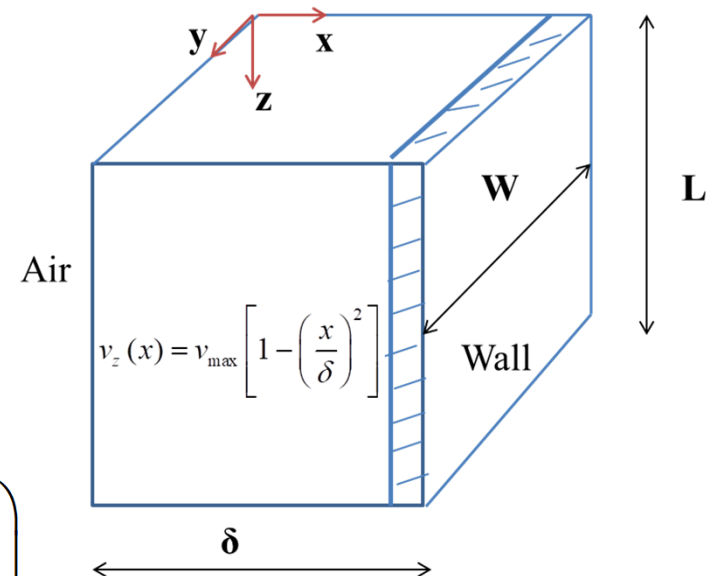
BC

1. At $z = 0$, $C_A = 0$

2. At $x = 0$, $C_A = C_{AO}$

3. At $x = \infty$, $C_A = 0$

$$\frac{C_A}{C_{AO}} = 1 - \operatorname{erf} \left(\frac{x}{\sqrt{4D_{AB} z / v_{\max}}} \right)$$

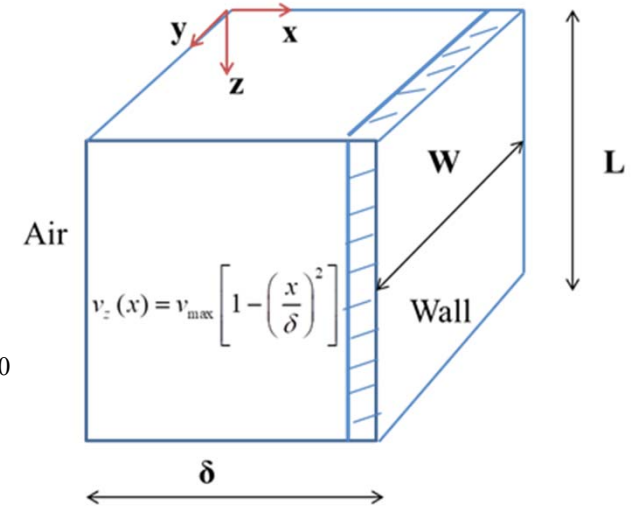


The total mass transfer rate can be calculated by integration of the local mass flux at ($x = 0$)

The local mass transfer rate at $x = 0$

$$N_{Ax}(z)_{x=0} = -D_{AB} \left. \frac{\partial C_A}{\partial x} \right|_{x=0} = C_{A0} D_{AB} \frac{\partial}{\partial x} \left[\operatorname{erf} \frac{x}{\sqrt{4D_{AB}z/v_{\max}}} \right]_{x=0}$$

$$N_{Ax}(z)_{x=0} = C_{A0} \sqrt{\frac{D_{AB} v_{\max}}{\pi z}}$$



Total moles of A transferred per unit time from the gas to the film is

$$W_A = \int_0^W \int_0^L N_{Ax_{x=0}} dz dy = W C_{A0} \int_0^L \sqrt{\frac{D_{AB} v_{\max}}{\pi z}} dz = W L C_{A0} \sqrt{\frac{4 D_{AB} v_{\max}}{\pi L}}$$

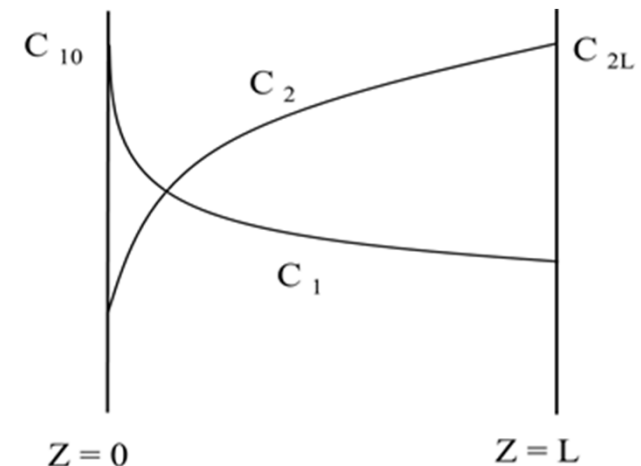
Total moles of A transferred per unit time from the gas to the film is

$$W_A = \int_0^W \int_0^L N_{Ax_{x=0}} dz dy = W C_{A0} \int_0^L \sqrt{\frac{D_{AB} v_{\max}}{\pi z}} dz = W L C_{A0} \sqrt{\frac{4 D_{AB} v_{\max}}{\pi L}}$$

The mass transfer rate is

- Directly proportional to the square root of the diffusivity
- Inversely proportional to the square root of the exposure time, $L/v_{\max}$

Imagine a film of stagnant liquid in which a chemical species of concentration C_2 is dissolved. At the surface $z = 0$, in contact with a solid wall, a concentration C_{10} of another species is maintained. The thickness of the film is L , over which C_1 changes from a concentration of C_{10} to 0. and it can be assumed that some external mechanism is present that permits us to maintain C_{10} at $z = 0$ and C_{2L} at $z = L$. Species 1 and 2 will diffuse and profiles $C_1(z)$ and $C_2(z)$ will be established at steady state



Now further suppose that species 1 and 2 react at a rate $R_{12} = -k C_1 C_2$ (R_{12} is the rate of disappearance of 1 and 2). This reaction occurs homogeneously throughout the film. Species 1 and 2 diffuse through the film according to different diffusion coefficients, denoted by D_1 and D_2 respectively. Derive (do not solve) the governing equations (for one-dimensional diffusion only process) for the two species and state clearly the boundary conditions that reflect the physics of the process.

For 1D steady diffusion, the governing equations are -

$$0 = D_1 \frac{d^2 C_1}{d z^2} - k C_1 C_2 \dots\dots\dots(1)$$

$$0 = D_2 \frac{d^2 C_2}{d z^2} - k C_1 C_2 \dots\dots\dots(1)$$

With boundary conditions that reflect the following physical ideas -

At $z = 0$, C_1 is maintained at C_{10} and species 2 does not cross the plane at $z=0$

$$C_1 = C_{10} \quad \text{at } z = 0$$

$$\frac{d C_2}{d z} = 0 \quad \text{at } z = 0$$

At $z = L$, C_2 is maintained at C_{2L} and species 1 is depleted by reaction before reaching that plane

$$C_2 = C_{2L} \quad \text{at } z = L$$

$$C_1 = 0 \quad \text{at } z = L$$

Governing equations (1) and (2) are non-linear in the concentration C_1 and C_2 .

An analytical solution is not possible.

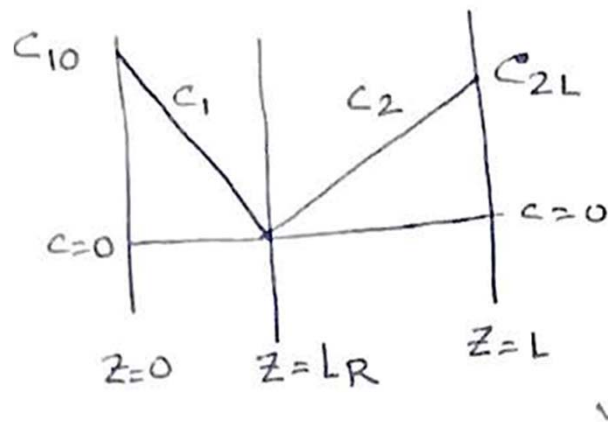
An analytical solution of these coupled nonlinear differential equations is not possible. However, one can make an approximation if it is assumed that the reaction rate is very fast i.e., when the diffusing molecules meet, they react and disappear immediately. Let the plane at which they meet, $z = L_R$ is called the "reaction front". Derive analytical expressions for the concentration distributions of the two species for this case (in terms of L_R and other constants) within the respective domains of their existence and sketch the profiles.

Obtain an expression for the location of this "reaction front" i.e. an expression for L_R .

If the reaction rate is very fast

When the molecules meet they react and disappear at a plane ($Z = L_R$) - the reaction front

If the reaction is sufficiently fast, the profiles of C_1 and C_2 will appear as follows:



It is assumed that there is no excess of species 2

In the region $0 \leq Z \leq L_R$ there is no species 2

$$\frac{d^2 C_1}{d z^2} = 0$$

with $C_1 = C_{10}$ at $z = 0$

and $C_1 = 0$ at $z = L_R$

By a similar argument, the C_2 profile satisfies

$$\frac{d^2 C_2}{dz^2} = 0$$

$$\text{with } C_2 = C_{2L} \text{ at } z = 0$$

$$\text{and } C_2 = 0 \text{ at } z = L_R$$

The solutions are

$$C_1 = a_1 + b_1 z$$

$$C_2 = a_2 + b_2 z$$

$$a_1 = C_{10} ; \quad b_1 = -\frac{C_{10}}{L_R} \quad a_2 = -\frac{C_{2L} L_R}{L - L_R} \quad b_2 = \frac{C_{2L}}{L - L_R}$$

But L_R is unknown

Additional physical statement: Since 1 mole of species 1 reacts with 1 mole of 2, at steady state

$$N_1 = -N_2$$

$$-D_1 \frac{dC_1}{dZ} = D_2 \frac{dC_2}{dZ}$$

$$-D_1 b_1 = D_2 b_2$$

$$\frac{C_{10}D_1}{L_R} = \frac{C_{2L}D_2}{L - L_R} \quad \Rightarrow \quad L_R = \frac{L}{\left(1 + \frac{C_{2L}D_2}{C_{10}D_1}\right)}$$

$$\text{and} \quad N_1 = \frac{C_{10}D_1}{L} \left(1 + \frac{C_{2L}D_2}{C_{10}D_1}\right)$$

The rate of transfer of 1 into the film is enhanced by the reaction, i.e. the chemical reaction may serve to enhance the rate of absorption of a gas by a liquid.

- Design of absorbers
- Biomedical devices

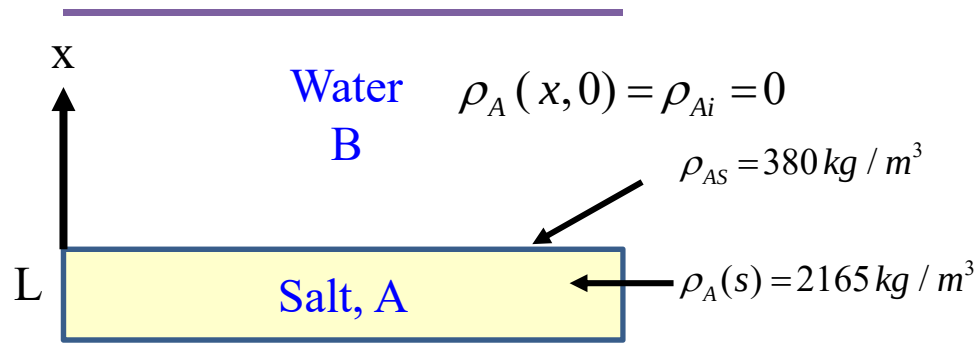
A slab of salt of thickness L is used to support a deep layer of water. The salt dissolves in the water, maintaining a fixed mass density, ρ_{As} (Kg/m^3) at the water-salt interface. If the salt density in water is initially zero, how does this density vary with position and time after contact between the solid salt and the water has been made?

What is the surface recession rate dL/dt and how does the surface recession vary with time?

If the mass density of the solid salt is $\rho_A(S) = 2165 \text{ (Kg/m}^3\text{)}$ and the density in the solution at the surface is $\rho_{As} = 380 \text{ Kg/m}^3$, by how much the surface recede after 24 h?

The salt-water diffusion coefficient is $D_{AB} = 1.2 \times 10^{-9} \text{ m}^2/\text{s}$.

$$\left(\frac{\partial \rho_A}{\partial t} + v_x \frac{\partial \rho_A}{\partial x} + v_y \frac{\partial \rho_A}{\partial y} + v_z \frac{\partial \rho_A}{\partial z} \right) = D_{AB} \left[\frac{\partial^2 \rho_A}{\partial x^2} + \frac{\partial^2 \rho_A}{\partial y^2} + \frac{\partial^2 \rho_A}{\partial z^2} \right] + r_A$$



$$\frac{\partial \rho_A}{\partial t} = D_{AB} \frac{\partial^2 \rho_A}{\partial x^2}$$

$$IC: \quad \rho_A(x,0) = \rho_{Ai} = 0$$

$$BC: \quad \rho_A(\infty, t) = 0 \quad \rho_A(0, t) = \rho_{AS}$$

Solution

$$\frac{\rho_A(x,t)}{\rho_{AS}} = 1 - \operatorname{erf} \left[\frac{x}{2(D_{AB} t)^{1/2}} \right]$$

Performing a mass balance around the slab

$$-\dot{M}_{A,OUT} = \dot{M}_{A,Stored}$$

$$\frac{\rho_A(x,t)}{\rho_{AS}} = 1 - \operatorname{erf} \left[\frac{x}{2(D_{AB}t)^{1/2}} \right]$$

$$-\dot{M}_{A,OUT} = \dot{M}_{A,Stored}$$

Based on unit surface area

$$-n_{AS}'' A = \frac{d}{dt}(\rho_A(s)L)A$$

$$-n_{AS}'' = \frac{D_{AB} \rho_{AS}}{(\pi D_{AB} t)^{1/2}}$$

$$\frac{D_{AB} \rho_{AS}}{(\pi D_{AB} t)^{1/2}} = \rho_A(s) \frac{dL}{dt}$$

$$\left. \frac{d}{dx} \left\{ \operatorname{erf} \frac{x}{2(D_{AB}t)^{1/2}} \right\} \right|_{x=0} = \frac{1}{(\pi D_{AB} t)^{1/2}}$$

$$\Delta L = -\frac{2\rho_{AS}}{\rho_A(s)} \left(\frac{D_{AB}t}{\pi} \right)^{1/2}$$

$$= -2 \times \frac{380 \text{ kg/m}^3}{2165 \text{ kg/m}^3} \left(\frac{1.2 \times 10^{-9} \text{ m}^2/\text{s} \times 24 \text{ h} \times 3600 \frac{\text{s}}{\text{h}}}{\pi} \right)^{1/2}$$

$$= 2.02 \text{ mm}$$